Research Dissertation (MSc)

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Research Dissertation (MSc) (A36382)

Short Title: Research Dissertation (MSc)
Faculty Science and Computing
Department Science
Cluster Placement and Projects
Author KGRENNAN
Credits 30 **Level:** Postgraduate

Description of Module / Aims

This module will develop the learner s ability to access relevant scientific information from the primary literature, collate the information and write an overview of a chosen research topic. Projects related to this research topic will be carried out under the supervision of individual members of the academic staff, along with an internal supervisor at the workplace. Projects will be linked to a relevant industrial problem at the student s workplace. The student will present their research findings in the form of a research dissertation and also at a presentation in a formal research environment.

Programmes

DISS-0096	MSc in Analytical Science with Quality Management (WD_SANSC_R)	1 / 3 / M

Indicative Content

- Literature search strategies, including the use of online databases to effectively research the scientific literature
- Preparation of a literature review and critical evaluation of the literature related to a scientific research topic
- Use of specialised referencing manager packages
- Technical writing skills & designing report templates
- Writing science correctly
- Planning and design of a scientific project in the context of a literature review
- Interpretation and critical evaluation of results in the context of the relevant scientific literature
- Quality of written report on project
- Preparation and delivery of a presentation suitable for a technical audience

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Evaluate information contained in the primary literature pertaining to a given research topic and determine its relative importance.
2. Construct a cohesive and fully referenced literature review report on a given research topic that conforms to a high standard of technical writing.
3. Appraise the parameters and components of an area of study, presenting these in the form of a hypothesis or problem statement and describing a methodology for investigation.
4. Develop powers of analysis and synthesis in the exploration and resolution of problems.
5. Assess and defend their work, its conclusions and recommendations in written form.
6. Design a well-illustrated presentation to summarise the research project suitable for dissemination to a technical audience.

Learning and Teaching Methods

- Dissertation.
- Presentation.

Assessment Methods

	Weighing	Outcomes Assessed
Dissertation	70%	1,2,3,4,5
Continuous Assessment	30%	6
Presentation	30%	6

Assessment Criteria

<40%: Insufficient level of research work performed in terms of quality and quantity of research output. Inadequate standard and knowledge of research topic evident through poor report writing and oral presentation skills.

40%-59%: Limited consideration of most points and concepts addressed. Descriptive level of discussion, showing lack of evidence of the underpinning knowledge. Some attempt at application of theory to practice.

60%-69%: Addressing and analysing the main points and showing evidence of underpinning knowledge. Some evidence of evaluation and synthesis of the relevant issues and theory to practice integration.

70%-100%: Good level of critical analysis, originality of thought and a comprehensive knowledge base. Good critical evaluation and synthesis of relevant issues, showing an ability to integrate theory and practice.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Online Delivery		24
Independent Learning		786

Essential Material

- "Academic Search Complete database - WIT library databases." http://library.wit.ie/Resources/databases
- "Emerald database - WIT library databases." http://library.wit.ie/Resources/databases
- "Research Dissertation module Moodle page." http://moodle.wit.ie
- "ScienceDirect database - WIT library databases." http://library.wit.ie/Resources/databases
- "WIT library catalogue." http://library.wit.ie/
- "Web of Science database - WIT library databases." http://library.wit.ie/Resources/databases

Supplementary Material

- Alley, M. _The craft of scientific writing_. New York: Prentice-Hall, 2009.
- Carter, M. _Designing science presentations_. San Diego, CA: Elsevier, 2013.
- Lebrun,, J.L. _Scientific writing 2.0: a reader's and writer's guide_. Hackensack, N.J.: New World Scientific, 2011.
- Marder, M.P. _Research methods for science_. Cambridge: Cambridge University Press, 2011.

- Neville, C. _Complete guide to referencing and avoiding plagiarism_. Maidenhead, NY: Open University Press, 2010.